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CSA-103

Examination –May- 2024

B.Tech. I/II Sem: CS, IT, AI, CSE(BC), IoT

Problem Solving using Data Structures

Time :2Hrs

Max. Marks : 60
Min. Marks :19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c &d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a & b – Objective type/Fill in the blanks/One sentence answer.
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)	Which of the following functions is used in C to allocate memory dynamically? (i) malloc() (ii) scanf() (iii) printf() (iv) main()	02	4	1
	(b)	Which of the following best describes the top-down design approach? (i) Starting with specific details and moving to a general overview. (ii) Focusing on improving the efficiency of the solution. (iii) Breaking down a complex problem into smaller, more manageable sub-problems. (iv) Integrating multiple existing solutions to form a complete system.	02	4	1
	(c)	What are some applications of Data Structures?	03	2	1
		OR			
		How are the elements of a 2D array stored in the memory?	03	2	1
	(d)	What do you mean by an algorithm? Write the criteria and characteristics of an algorithm.	05	2	1
		OR			
		Write a program to Insert a new item at beginning, end and middle position within an array	05	3	1
Q.2	(a)	What is the time complexity of adding two polynomials represented by linked lists? (i) $O(1)$ (ii) $O(n)$ *(iii)* $O(n\log n)$ *(iv)* $O(n^2)$	02	3	2
	(b)	In a singly linked list, each node contains: (i) Data and a pointer to the next node. (ii) Only data. (iii) Data and two pointers to the next and previous nodes. (iv) Data and a pointer to the previous node.	02	3	2
	(c)	What are some applications of Linked List?	03	2	2
		OR			
		If data is stored in the form of single linked list. Which searching method is most suitable for this type of data structure?	03	3	2
	(d)	What are Circular, Doubly and Singly linked list? Write a program to insert a node in circular Linked list.	05	4	2
		OR			
		Write a program to create a linked list data structure using pointers in C?	05	4	2

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Q.3	(a)	Which of the following operations is not a basic stack operation? (i) Push (ii) Pop (iii) Peek (iv) Enqueue	02	3	3	
	(b)	In a circular queue, how is the "full" condition typically determined? (i) When front equals rear. (ii) When (rear + 1) % size equals front. (iii) When front is greater than rear. (iv) When the queue is empty.	02	3	3	
	(c)	What is Stack? Where do we use Stack data structure.	03	2	3	
		OR				
		List some applications of queue Data Structure.	03	2	3	
	(d)	Explain how to evaluate arithmetic expression using stack?	05	2	3	
		OR				
		Write an algorithm to insert and delete an element in circular queue.	05	4	3	
Q.4	(a)	Which of the following is not a type of tree traversal? (i) Pre-order (ii) In-order (iii) Post-order (iv) Side-order	02	3	4	
	(b)	Which graph traversal technique uses a queue data structure? (i) Depth-First Traversal (ii) Breadth-First Traversal (iii) In-order Traversal (iv) Pre-order Traversal	02	3	4	
	(c)	Differentiate between full binary tree and complete binary tree.	03	3	4	
		OR				
		Differentiate between Depth First Search and Breadth First Search.	03	3	4	
	(d)	Construct AVL search tree by inserting the following elements in order of their occurrence.65,5,38,24,18,116,92,82,48	05	4	4	
		OR				
		Explain the various representation of graph with example in detail?	05	2	4	
Q.5	(a)	Which sorting algorithm works by selecting the smallest (or largest) element from an unsorted list and swapping it with the element at the beginning of the list? (i) Bubble sort (ii) Insertion sort (iii) Selection sort (iv) Merge sort	02	3	5	
	(b)	Which of the following is a method to handle collisions in a hash table? (i) Quick sort (ii) Binary search (iii) Chaining (iv) Depth-first search	02	3	5	
	(c)	What do you mean by hashing?	03	3	5	
		OR				
		Differentiate between internal sorting and external sorting.	03	3	5	
	(d)	Sort the following list using quick sort 36,95,42,12,08,66,72,55	05	4	5	
		OR				
		Implement Binary Search Algorithm using Recursive Approach	05	4	5	

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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